INCH-POUND

MIL-PRF-26392D (USAF)
26 December 1996
SUPERSEDING
MIL-D-26392C (USAF)
28 OCTOBER 1994

PERFORMANCE SPECIFICATION

DUMMY CONVERTERS, LIQUID OXYGEN INDICATOR SYSTEM,

10-LITER CRU-23/A AND 25-LITER CRU-24/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of dummy converters (electrical capacitors) having a capacitance equivalent to an empty 10-liter and 25-liter liquid oxygen converter.

1.2 Classification. The dummy converters are classified as follows (see 6.4):

Type	Designation	Liters
I	CRU-23/A	10
II	CRU-24/A	25

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of this document are those listed in the specific issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Suite 1AE1-101A, Tinker AFB, OK 73145-3036 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660
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SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-C-25516 Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification For

(Unless otherwise indicated, copies of the above specification are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the dummy converter shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meet or exceed the operational and maintenance requirements, and promote economically advantageous life cycle costs.

3.3 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition or other conditions that may be encountered during operational use or storage.

3.4 Interface.

3.4.1 Dimensions. The dummy converter case, mounting plate, and electrical connectors shall conform to the dimensions in Figure 1.

3.4.2 Electrical connectors. The high impedance sensing element lead shall be terminated in a connector conforming to MIL-C-25516, B polarity. The low impedance lead shall be terminated in a connector conforming to MIL-C-25516, E polarity. The connectors shall provide for waterproof connections and have dust caps to prevent the entrance of foreign particles. The dust caps shall be tethered to prevent movement or loss when not installed on the connectors.

3.5 PERFORMANCE

3.5.1 Capacitor. Type I dummy converter capacitor shall have an electrical capacitance of 123.5 ± 1.0 picofarads (pF), and the Type II shall have an electrical capacitance of 303.5 ± 2.5 pF. Capacitance shall remain constant at temperatures between -65° and 260°F. The rated voltage of the capacitors shall be 110V.

3.5.2 Case. The case shall provide a moisture-tight seal.

3.5.3 Reliability. When specified (see 6.2), the dummy converter shall have a mean time between failures (MTBF) of not less than 300 hours, or a minimum acceptable MTBF of 200 hours at a reliability of .9933 for a 2-hour mission.

3.5.4 Maintainability. The total corrective maintenance time for field level expected failures shall not exceed 1 hour.

3.5.5 Weight. The weight of the complete dummy converter shall not exceed 0.25 pounds.

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3.5.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one dummy converter and consist of all examinations and tests specified (see 4.6).

4.3 Conformance inspection. Conformance inspections shall include the tests specified (see 4.3.1 and 4.3.2).

4.3.1 Individual tests. Each dummy converter shall be subjected to the following examination and tests.

a. Examination of product (see 4.6.1).

b. Capacitance at room temperature (see 4.6.2).

4.3.2 Sample tests. One percent of each order (a minimum of one dummy converter), shall be subjected to the following tests.

a. Capacitance at low temperature (see 4.6.3).

b. Capacitance at high temperature (see 4.6.4).

c. Immersion (leakage) (see 4.6.5).

4.4 Inspection conditions. Unless otherwise specified, all tests shall be performed at ambient temperature and pressure, and at a maximum relative humidity of 80 percent.

4.5 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE I: Requirements cross-reference matrix

REQUIREMENT	VERIFICATION
3.1	4.2
3.3	4.6.1
3.4.1	4.6.1
3.4.2	4.6.1
3.5.1	4.6.2, 4.6.3, 4.6.4
3.5.2	4.6.5
3.5.3	4.6.6
3.5.4	4.6.7
3.5.5	4.6.1
3.5.6	4.6.1

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4.6 Tests.

4.6.1 Examination of product. Each dummy converter shall be examined to determine conformance to materials, weight, and dimensions.

4.6.2 Capacitance at room temperature. Using a capacitance measuring instrument, the capacitance at room temperature shall be 123.5 ± 1.0 pF for Type I dummy converters and 303.5 ± 2.5 pF for Type II dummy converters.

4.6.3 Capacitance of low temperature. The dummy converter shall be cold soaked at -65°F for at least 8 hours. While still at this temperature, the capacitance shall be 123.5 ± 1.0 pF for Type I dummy converters and 303.5 ± 2.5 pF for Type II dummy converters.

4.6.4 Capacitance at high temperature. The dummy converter shall be maintained at a minimum temperature of 260°F for at least 8 hours. While still at this temperature, the capacitance shall be measured and shall be 123.5 ± 1.0 pF for Type I dummy converters and 303.5 ± 2.5 pF for Type II dummy converters.

4.6.5 Immersion (leakage). The dummy converter, in normal operating configuration, shall be heated to 113° ± 5°F and maintained at this temperature for a maximum of 2 hours. The dummy converter then shall be immersed in water at 64° ± 9°F to a depth of 36 ± 5 inches for 2 hours. Any evidence of water penetration into the dummy converter shall be a cause for rejection.

4.6.6 Reliability. This test shall be conducted with sufficient duration to obtain a 90% confidence that an MTBF of 300 hours has been achieved. This test may also be used to demonstrate an MTBF of 200 hours and a 99.33% probability of completing a 2-hour mission. The dummy converter shall be stabilized at a temperature of -65°F, then the temperature shall be raised to 260°F and held for at least 2 hours. After 2 hours, the temperature shall be reduced to -65°F. The cycling time between temperature extremes shall not exceed 40 minutes. The dummy converter shall be subjected to at least four of these temperature cycles every 24 hours. The dummy converter shall be subjected to vibration at a frequency between 20 and 60 Hz and an applied acceleration of ±2g for at least 10 minutes during each hour of this test. The rated voltage within ±2% shall be applied continuously during this test. The dummy converter shall be checked at least once every 24 hours at each temperature extreme to determine if a failure has occurred. No parts shall be replaced as preventive maintenance during reliability testing. A failure shall be identified as any malfunction that causes a failure in the capacitance value of the dummy converter.

4.6.7 Maintainability. Completion of corrective maintenance tasks within 1 hour (field level only) for expected failures shall demonstrate compliance with the maintainability requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The dummy converters covered by this specification are intended to be used in F-111, F-14, F-15, and B-52 aircraft when one of the converters is not installed.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. Item identification.

d. Type of dummy converter (see 1.2).

e. When first article is required (see 3.1).

f. When reliability testing is required (see 3.5.3).

g. Data required (see Section 4).

h. Packaging requirements (see 5.1).

6.3 Cross reference. This specification supersedes MIL-D-26393. All items that conform to MIL-D-26393 are interchangeable with Type II dummy converters, PIN M26392-2.

6.4 Part or Identifying Number (PIN). The PIN is as follows:

PIN	Type	Designation	Liters
M26392-1	I	CRU-23/A	10
M26392-2	II	CRU-24/A	25

6.5 Subject term (key word) listing.

Breathing oxygen
Capacitance
Capacitor
Case
Interchangeability
Maintainability
Reliability

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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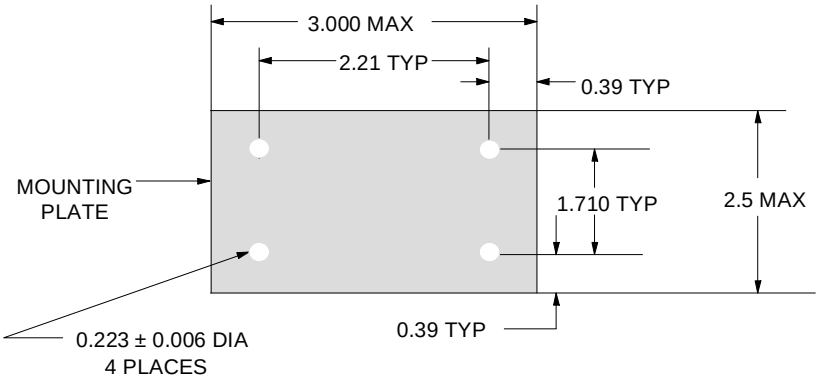

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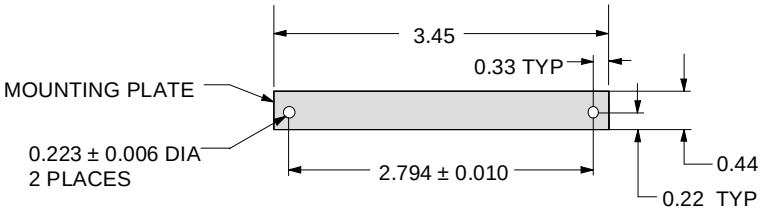
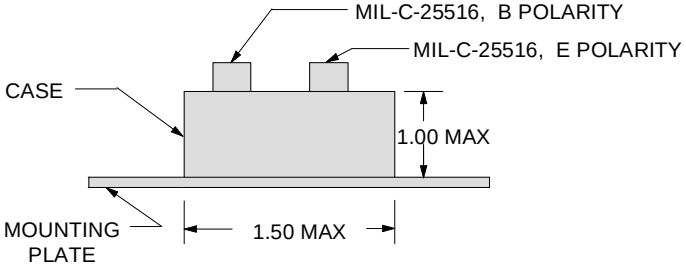


TOP VIEW TYPE I
(CASE NOT SHOWN)



SIDE VIEW TYPE I
AND TYPE II

NOTES: 1. DIMENSIONS IN INCHES

2. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ± 0.030 FOR .XX
± 0.05 FOR

FIGURE 1. Dimensions for case and mounting plate, Type I and Type II

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** **MIL-PRF-26392D**	**2. DOCUMENT DATE (YYMMDD)** **961226**

3. DOCUMENT TITLE
DUMMY CONVERTERS, LIQUID OXYGEN INDICATOR SYSTEM 10-LITER CRU-23/A AND 25-LITER CRU-24/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED (YYMMDD)**

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c. ADDRESS ***(Include Zip Code)*** ***OC-ALC/TICLA*** ***3001 Staff Drive Suite 1AE1-101A*** ***Tinker AFB, OK 73145-3036***	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 756-2340/2343 DSN 289-2340/2343 2340/2343**

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